

Supplementary Material

Corrected Statements and Proofs for the Revised Long-Memory Limit Theorems

Companion to `revised_theorems.tex`; machine-checked items verified in the Lean 4 repository at closure commit 16715db

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This supplement gives, for each substantive correction identified in the audit of 19-AOS1825.pdf: the original claim, the error (with the counterexample where one exists), the corrected statement, and a proof or proof sketch. Section 3 then proves in full the honest-rate drift proposition underlying the revised main theorem. Full written chains are in `direct_proofs/01–24`; per-item machine status is in `paper_revision/theorem_inventory.md`. Status tags: [Lean] = completely formalized; [written] = written mathematical proof.

1 Corrections to the main theorems

1.1 Critical-regime variance (Theorem 3.3(ii), via S.3.2)

Original. In the critical regime $H(t) = \frac{3}{4}$ the limiting variance is expressed through the kernel center value. **Error.** The derivation of the diagonal contribution ε_4 in S.3.2 replaces $f^2(i/N)f^2(j/N)$ by $f^4(0)$ for $|i - j|$ small, which is not summable to $f^4(0)$; the correct diagonal sum tends to $\int f^2$, and the Hermite rank-2 factor $2!$ together with the squared denominator of g was mishandled. **Corrected.**

$$\text{Var} \longrightarrow \frac{9}{16} \int \omega_r(x)^2 dx.$$

Proof sketch. The rank-2 diagonal term of the weighted double sum carries the factor 2; the diagonal sums $\sum_i K_i^2 f^2(i/N)^2$ converge to $\int \omega_r^2 f^2$ -type integrals which, at the critical exponent, pin the constant at $9/16$. Full chain: files 03/13.

[written]

1.2 Long-memory sign (Theorem 3.3(iii)–3.4)

Original. The inverted estimator converges to $+Q$. **Error.** The inversion $x \mapsto (c_\sigma - x)/(2 \log n)$ carries a negative slope, so the limit reflects: $-Q$. This is substantive: a legal nonnegative equivalent-kernel instance has third moments of Q and $-Q$ of opposite sign, so the two laws differ. **Corrected.** $2S_n^\psi \log n (\hat{H}_n - h) \Rightarrow -Q$ (Theorem 1 of the companion note). **Proof.** Expectation-centered limit for $\hat{G}$ with the reflection through the decreasing calibration; the sign is retained throughout the transport. The full statement is machine-checked.

[Lean]

1.3 Optimal-bandwidth biased means (Corollary 3.5)

Original. Both $\widehat{G}$ and $\widehat{H}$ limits carry mean $R(t)$. **Error.** The two means have opposite signs. **Corrected.** With $b/b_0 \rightarrow 1$, $b_0 = (n^d \log^2 n)^{-1/(2p+d)}$:

$$\text{mean}(\widehat{G}) = -2R(t), \quad \text{mean}(\widehat{H}) = 2R(t), \quad \text{variance} = \xi^2(t) \text{ in both;}$$

if $b/b_0 \rightarrow c$, multiply the means by c^p and the variances by c^{-d} . R may take either sign; $R = 0$ requires the o -term from the bias expansion, not an equivalence with 0. **Proof sketch.** The smoothing bias enters $\mathbb{E}\widehat{G}$ with the sign fixed by the second-order expansion; the inversion flips it. Files 13/21.

[written]; conditional Lean chain connected

1.4 Growing-lag remainder (Theorem 8.2(iv))

Original. A stronger remainder is claimed. **Corrected.** For effective interior differences with $b \log n \rightarrow 0$,

$$\text{error} = O\left(k^{-2} + \frac{k \log n}{n} + \left(\frac{k}{n}\right)^\psi\right) \quad (q = 1), \quad O\left(k^{-2} + \frac{k \log n}{n}\right) \quad (q = 2),$$

and the finite-lag error must be retained. **Reason.** Cross terms between lag growth, mesh resolution and the Hölder exponent cannot be dropped. Files 12/18.

[written]; interior pieces Lean

1.5 Local-polynomial bias (Lemma 8.4)

Original. An integral-region/quadrature argument. **Error.** The quadrature over the integration region is invalid. **Corrected.** Define $M_{kl} = \sum_i K_i A_{ik} A_{il}$ and $w_i = K_i \sum_l (M^{-1})_{0l} A_{il}$. **Proof.** Invertibility of M gives $\sum_i w_i A_{ik} = \delta_{k0}$ exactly; hence for $F_i = \beta_0 + \sum_{k \geq 1} \beta_k A_{ik} + r_i$ with $|r_i| \leq B$ on the support of K ,

$$\left| \sum_i w_i F_i - \beta_0 \right| \leq \sum_i |w_i| B = O(b^p),$$

using the uniform L^1 weight bound. No $(nb)^{-\min(2,p)}$ term is needed.

[Lean](designs, weights, equivalent kernel, bias leading term)

1.6 Vector Z limits (Lemma 8.5)

Original. Limits stated for Z itself; dimension a taken as d . **Corrections.** State the limits for $Z - \mathbb{E}Z$; keeping Z requires separately proving $\mathbb{E}Z \rightarrow 0$ after normalization — the bound $\mathbb{E}Z = O((nb)^{d/2} \rho)$ alone does not suffice. The dimension is the polynomial basis dimension S , not d ; kernel conditions added.

expectation-bias leading terms [Lean]; three Z -limits [written]

2 Corrections to the supplementary lemmas

2.1 S.1.1 (Fourier adjacent differences)

Error. As a complex Lebesgue integral the claim fails: near 0 the imaginary part behaves like $-ikx/|x|^{2+\delta}$, which is not absolutely integrable. **Correction.** Replace $1 - e^{ikx}$ by $1 - \cos(kx)$, or define the principal value explicitly; the scaling and mean-value steps then give the adjacent-difference bounds.

[written](repair); not formalized

2.2 S.2.1 (g bounded away from zero)

Error. Insufficient hypotheses: at $h = 0$ the quantity vanishes, and for $q = 2$, $H \rightarrow 1$ no uniform positive lower bound exists. **Correction.** Require $h \neq 0$ and $H \in [\gamma, 1 - \gamma]$, the latter following from [A1] rather than [A3]; positivity, continuity and compactness then give the uniform lower bound.

repaired 1-d version [Lean]

2.3 S.2.2 / S.3.1 (leading singular coefficient)

Error. The printed c_h misses a factor $\frac{1}{2}$. **Correction.** $c_h = h(2h - 1)$, i.e. $g(H, u, h) = (c_h + o(1))|u|^{-\psi}$ with $\psi = 2 - 2H$ (frozen H ; all derivatives acting on the distance in the principal term, H -derivatives in the remainder). The $o(1)$ must not be strengthened to $O(b \log b)$.

instantiated and machine-verified inside the closure chain [Lean]

2.4 S.2.3 (mixed derivatives, near-diagonal)

Error. The $q = 1$ near-diagonal remainder $O(r \log r)$ admits a lawful linear- H counterexample.

Correction. Add the smooth background term of relative size r^ψ : the corrected bound retains the $x^{H(x)+H(y)-q}$ -times-log-power background near 0 and normalizes on the actual grid.

[written]; grid normalization pieces [Lean]

2.5 S.3.2 (weighted power sums / cycle products)

Corrected jointly with §1.1 above (center value $\int f^2$; 2!; denominator). The long-memory side — identification of the cycle products with the spectrum of $B = K^{1/2} M_\omega K^{1/2}$ for all orders $k \geq 2$ — is machine-checked.

long-memory identification [Lean]

2.6 S.5.1 (spatial average variance of I'_1)

Error. The far-distance count misses a factor m : with $H \equiv 7/8$, $q = p = 1$, $b = n^{-1/2}$ one has $\text{Var}(I_1) \approx \log^2 n \cdot n^{-1/2}$ against the claimed $O(\log^2 n \cdot n^{-3/4})$. **Correction.** The count is $m^{2\psi-1} n^{-2\psi}$ times the lag sum; merging weights at $m \approx 1/b$ gives $\text{Var}(I_1) \leq C \log^2 n \cdot T(n)$, which the original bandwidth condition still absorbs into the target term — the main scale rates are preserved.

[written]; the $q = 1$ scale input it feeds is discharged in Lean

2.7 S.6.1 (irregular-grid covariance)

Error. As printed (unnormalized covariance) the claim fails already for Brownian motion. **Correction.** Either estimate $\text{Cov}(W_n(t), W_n(s))$ with $W_n(t) = \sigma^{-1} n^{H(t)} \Delta^q X(t)$, or restore $\sigma^2 n^{-H(t)-H(s)}$ on both sides of the original.

scalar counterexample [Lean]; full instantiation open

2.8 S.7.2 (dimension scope)

No counterexample; scope remark: the $d = 2, 3$ “minimax optimality” assertion has no corresponding lower bound in the manuscript itself, so it cannot be certified even after repairing the upper-bound route.

[written](route only)

3 Proof of the honest-rate drift proposition

This section proves Proposition 2 of the companion note in full. Work under (R1)–(R6) of the companion note: $\delta_n = n^{-h}$, $S_n = n^{1-h}$, $\psi = 2 - 2h \in (0, \frac{1}{2})$, $c_\sigma = \mathbb{E} \log Z^2$, $\widehat{G}_n$ the unit-weight log-square statistic, $\widetilde{H}_n = (c_\sigma - \widehat{G}_n)/(2 \log n)$, $\widehat{H}_n = T(\widetilde{H}_n)$ clipped to $[0, 1]$.

Proposition 1. $2S_n^\psi \log n |\mathbb{E} \widehat{H}_n - h| \rightarrow 0$, with the three-term bound of the companion note; the second term vanishes exactly under the b -band $(1 - h)^2 < 1 - b$, the other two unconditionally.

Proof. Step 1 (exact expectation identity). For each active row j the coefficient-feature combination satisfies, by the stride identity, the norm decomposition

$$\log \|\text{combo}_j\|^2 = -2H_j \log n + \log \|V_j\|^2,$$

where $H_j = f(s_j)$ is the left-endpoint Hölder value and V_j the normalized varying increment. The Gaussian expectation formula gives

$$\mathbb{E} \widehat{G}_n = \sum_j w_j (-2H_j \log n + \log \|V_j\|^2) + c_\sigma, \quad \sum_j w_j = 1,$$

whence, with c_σ pinned,

$$\mathbb{E} \widetilde{H}_n - h = -\left(\sum_j w_j H_j - h\right) - \frac{1}{2 \log n} \sum_j w_j \log \|V_j\|^2. \quad (1)$$

Step 2 (deterministic side). (i) *Weight contraction.* On the active window $|s_j - t| < \delta_n$, so Hölder regularity gives $|H_j - h| \leq D(1 + M)\delta_n$, and $|\sum_j w_j H_j - h| \leq \sum_j |w_j| |H_j - h| \leq C_{\text{st}} \delta_n^p$ with $p \geq 1$ (the sharper exponent uses the class regularity; $p = 1$ suffices). (ii) *Second-order increment log rate.* Write $V_j = F_k(s_j + \ell)$ -type normalized varying increment with $|k - h| \leq K\ell$, $\ell = 1/n$, $K = D(1 + M)$. The exact kernel identity

$$2\langle \text{frozen, mismatch} \rangle = \Lambda \Phi(m) - \Phi(h), \quad \Phi(q) = (s + \ell)^{2q} - s^{2q} + \ell^{2q}, \quad m = \frac{h+k}{2},$$

with $\Lambda - 1 = -\frac{1}{2} \|F_k(1) - F_h(1)\|^2 = O((k - h)^2)$ (uniform parameter Lipschitz; no smoothness of the normalizing constant needed) and the mixed-power Lipschitz bound $|\Phi(m) - \Phi(h)| \leq C|k - h|\ell$ (valid since $2a > 1$ absorbs the logarithmic sensitivity) yields

$$|\|V_j\|^2 - 1| \leq C(1 + K + K^2 + K^3) \ell^{2-2b} = C n^{-(2-2b)},$$

hence $|\log \|V_j\|^2| \leq 2C n^{-(2-2b)} =: C_{\log} n^{-(2-2b)}$ once small, and $|\sum_j w_j \log \|V_j\|^2| \leq C_w C_{\log} n^{-(2-2b)}$ with $C_w = \sum_j |w_j|$ uniformly bounded.

Step 3 (random side; tail-based transfer). Let $m = \mathbb{E} \widetilde{H}_n$. Since $\mathbb{E} \widetilde{H}_n \rightarrow h \in (3/4, 1)$ (companion note, center-band discharge), eventually $m \in [d', 1 - d']$ with $d' = (1 - h)/2$. On $\{\widetilde{H}_n \notin [0, 1]\}$ we have $|\widetilde{H}_n - m| \geq d'$. The unit-weight variance bound and the normalized energy give

$$V_n := \text{Var}(\widetilde{H}_n) = \frac{\text{Var}(\widehat{G}_n)}{4 \log^2 n} \leq \frac{4 g_{\text{LSV}} U^2 C_2}{4 \log^2 n} S_n^{-2\psi} = O\left(\frac{n^{-4(1-h)^2}}{\log^2 n}\right),$$

and Chebyshev gives $\varepsilon_n := \mathbb{P}(|\widetilde{H}_n - m| \geq d') \leq V_n/d'^2$. Since $|T(x) - x| \leq |x - m|$ off $[0, 1]$ and $T(x) = x$ on $[0, 1]$,

$$|\mathbb{E} \widehat{H}_n - \mathbb{E} \widetilde{H}_n| \leq \mathbb{E}[|\widetilde{H}_n - m| \mathbf{1}_{\widetilde{H}_n \notin [0, 1]}] \leq V_n + \varepsilon_n \leq V_n(1 + d'^{-2}).$$

Step 4 (exponents). Combining $\mathbb{E}\hat{H}_n - h = [\mathbb{E}\tilde{H}_n - h] + [\mathbb{E}\hat{H}_n - \mathbb{E}\tilde{H}_n]$ with Steps 2–3,

$$2S_n^\psi \log n |\mathbb{E}\hat{H}_n - h| \leq \underbrace{C_1 \frac{\log n}{n^{h-2(1-h)^2}}}_{\text{smoothing}} + \underbrace{C_2 n^{-(2-2b)-2(1-h)^2}}_{\text{mismatch}} + \underbrace{C_3 \frac{1}{n^{2(1-h)^2} \log n}}_{\text{clipping}}.$$

Indeed: the smoothing term is $2S_n^\psi \log n C_{\text{st}} \delta_n^p$ with $S_n^\psi = n^{2(1-h)^2}$ and $\delta_n^p = n^{-hp}$, and $2(1-h)^2 - hp \leq 2(1-h)^2 - h < 0$ since $2(1-h)^2 < \frac{1}{8} < \frac{3}{4} < h$; the mismatch term is $S_n^\psi C_w C_{\log} n^{-(2-2b)}$ with exponent $2(1-h)^2 - (2-2b)$, negative exactly when $(1-h)^2 < 1-b$, i.e. (R4); the clipping term is $2S_n^\psi \log n V_n(1+d'^{-2}) = O(n^{2(1-h)^2} \log n \cdot n^{-4(1-h)^2} / \log^2 n)$, unconditionally vanishing. $\square$

Remark 2 (Sharpness). The mismatch rate is attained: the expansion of $\log \|V_j\|^2$ has leading coefficient $f'(s)s^{2h-1}[2hD'/D + 2h \log s + 1]$ evaluated at the window, generically nonzero, so no per-row expansion order reaches the rate n^{-h} and the manuscript’s original bias envelope at that rate fails for non-constant profiles. The window refutations are machine-checked; the lower-bound statement is derived by hand and registered.

4 Formalization cross-reference

Every [Lean] tag above is a theorem in the repository, axiom-checked within `{propext, Classical.choice, Quot.sound}`, zero sorry; the entry points are `actualQ1_knownScaleH_fullChain_honestRate` (main theorem), `honestRate_drift_tendsto_zero` (Proposition 1), and the audit theorems listed in the companion note’s provenance table. Automated re-verification: `verification/final_acceptance.py`.